Exploring Vertical Data Format: ECLAT

- ECLAT (Equivalence Class Transformation): A **depth-first search** algorithm using set intersection [Zaki et al. @KDD'97]
- Vertical format
- Properties of Tid-Lists
  - t(X) = t(Y): X and Y always happen together (e.g., t(ac} = t(d})
  - t(X) $\subset$ t(Y): transaction having X always has Y (e.g., t(ac) $\subset$ t(ce))
- Frequent patterns: vertical intersections
- Using diffset to accelerate mining
  - Only keep track of differences of tids
  - t(e) = {T_{10}, T_{20}, T_{30}}, t(ce) = {T_{10}, T_{30}} → Diffset (ce, e) = {T_{20}}

t(e) = {T_{10}, T_{20}, T_{30}};
t(a) = {T_{10}, T_{20}};
t(ae) = {T_{10}, T_{20}}

A transaction DB in Horizontal Data Format

Tid	Itemset
10	a, c, d, e
20	a, b, e
30	b, c, e

The transaction DB in Vertical Data Format

Item	TidList
a	10, 20
b	20, 30
c	10, 30
d	10
e	10, 20, 30